Uniform Two-Parameter Asymptotics for Lanczos Coefficients of Centered Uniform Convolutions

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Abstract: Let $\beta_N^{[M]}$ be the monic three-term recurrence coefficients of the M-fold convolution of the uniform distribution on $[-1/2, 1/2]$, whose density is the centered cardinal B-spline of order M. We prove that $\beta_N^{[M]}/M^2 = F(N/M) + o(1)$ uniformly as $N, M \to \infty$ with N/M in a compact subset of $(0, \infty)$, where $F(\alpha) = a_\alpha^2/4$ and a_α is the Mhaskar–Rakhmanov–Saff number of the Cramér rate function of the uniform law. The proof combines a saddlepoint estimate with two derivatives, uniform in the cardinal B-spline family, with the varying-weight theorem of Levin and Lubinsky. The profile F increases from 0 to $1/16$, and $F(\alpha) = \alpha/12 - \alpha^2/20 + O(\alpha^3)$; its first five Taylor coefficients coincide with the leading-degree coefficients of Milovanović's Conjecture 2, which concerns the different regime of fixed degree. All properties of the limiting objects, the endpoint lemma relative to the saddlepoint bound, the elementary conditions of the Levin–Lubinsky class, the reduction of the Mhaskar–Rakhmanov–Saff equations and the logic of the uniformity argument are certified in Lean 4.

Key words: cardinal B-splines, Lanczos coefficients, varying weights, Mhaskar–Rakhmanov–Saff numbers, saddlepoint approximation, Lean 4 verification

1. Introduction.

The centered cardinal B-spline w_M of order M is the density of the sum S_M of M independent random variables uniformly distributed on $[-1/2, 1/2]$. Its orthogonal polynomials were computed symbolically by Milovanović [6] and numerically by Gautschi [4]. Milovanović also formulated two conjectures on the monic recurrence coefficients $\beta_N^{[M]}$, both in the regime of fixed degree N and growing order M. For fixed M and $N \to \infty$ the classical theory applies: w_M is positive on $(-M/2, M/2)$, so Rakhmanov's theorem (see [2]) gives $\beta_N^{[M]} \to M^2/16$. The present note concerns the intermediate, two-parameter regime in which the degree and the order grow together. There the external field of the scaled weight is asymptotically M times the Cramér rate function I of the uniform law, and the coefficients are governed by the Mhaskar–Rakhmanov–Saff equation of I. The precise statement is Theorem 7.1.

The proof applies the varying-weight theory of Levin and Lubinsky [5]. Its genuinely specific ingredient is a saddlepoint estimate for $\log w_M$ with two derivatives, uniform in M (Lemma 4.1). We do not claim a rate of

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convergence, although the computations of Section 9 suggest an error of order $1/M$. Section 2 fixes the notation and the classical inputs. Section 3 studies the limit profile. Sections 4–6 prove the three lemmas, and Section 7 proves the theorem. Section 8 treats the profile near 0 and ∞, and Section 9 reports the Lean 4 certification and the numerical checks.

2. Preliminaries.

2.1. Notation and conventions.

Let $U_1, U_2, \dots$ be independent and uniformly distributed on $[-1/2, 1/2]$. For $M \in \mathbb{N}$ put

$$S_M = \sum_{\ell=1}^{M} U_\ell, \qquad Y_M = \frac{S_M}{M}, \qquad g_M(x) = M w_M(Mx), \qquad \varphi_M = -\frac{1}{M}\log g_M, \tag{1}$$

so that w_M is the density of S_M, supported on $[-M/2, M/2]$, g_M is the density of Y_M, supported on $[-1/2, 1/2]$, and φ_M is defined on $(-1/2, 1/2)$. The monic orthogonal polynomials $P_N^{[M]}$ of $w_M(x)\,\mathrm{d}x$ satisfy, by symmetry, $P_0^{[M]} = 1$, $P_1^{[M]}(x) = x$ and

$$P_{N+1}^{[M]}(x) = x P_N^{[M]}(x) - \beta_N^{[M]} P_{N-1}^{[M]}(x), \qquad N \geq 1. \tag{2}$$

Thus $\beta_N^{[M]} = \|P_N^{[M]}\|^2 / \|P_{N-1}^{[M]}\|^2$ is the square of the Nth off-diagonal entry of the orthonormal Jacobi matrix; this is Milovanović's indexing [6, (2)–(3)], and translating the weight to $[0, M]$ does not change it. The Laplace transform of the uniform law and its logarithm are

$$A(z) = \int_{-1/2}^{1/2} e^{zt}\,\mathrm{d}t = \frac{2\sinh(z/2)}{z}, \qquad \Lambda(s) = \log\frac{\sinh(s/2)}{s/2}, \qquad s \in \mathbb{R}, \tag{3}$$

with $A(0) = 1$ and $\Lambda(0) = 0$. One has $\int e^{zy} w_M(y)\,\mathrm{d}y = A(z)^M$, and the derivatives of Λ are

$$\Lambda'(s) = \frac{1}{2} + \frac{1}{e^s - 1} - \frac{1}{s}, \qquad \Lambda''(s) = \frac{1}{s^2} - \frac{1}{4\sinh^2(s/2)}, \qquad s \neq 0, \tag{4}$$

with $\Lambda'(0) = 0$ and $\Lambda''(0) = 1/12$. By Lemma 3.1 below, Λ' is an increasing bijection of $\mathbb{R}$ onto $(-1/2, 1/2)$. Its inverse is denoted $s(x)$, and we put $v(x) = \Lambda''(s(x))$ and $\kappa_j(x) = \Lambda^{(j)}(s(x))$. The Cramér rate function and its derivatives are

$$I(x) = \sup_{s\in\mathbb{R}}\{sx - \Lambda(s)\} = s(x)\,x - \Lambda(s(x)), \qquad I'(x) = s(x), \qquad I''(x) = \frac{1}{v(x)}, \qquad |x| < \frac{1}{2}. \tag{5}$$

For a function G on $[0, 1/2)$ we write $\mathcal{H}_G(a) = \frac{1}{\pi}\int_0^{\pi/2} G(a\sin\theta)\,\mathrm{d}\theta$, and we use

$$H(a)=\frac{1}{\pi}\int_0^{\pi/2} a\sin\theta\, I'(a\sin\theta)\,\mathrm{d}\theta=\frac{1}{\pi}\int_0^{a}\frac{xI'(x)}{\sqrt{a^2-x^2}}\,\mathrm{d}x,\qquad H_M(a)=\frac{1}{\pi}\int_0^{\pi/2} a\sin\theta\,\varphi_M'(a\sin\theta)\,\mathrm{d}\theta, \tag{6}$$

for $0\le a<1/2$. For $\alpha>0$ we define $a_\alpha\in(0,1/2)$ by $H(a_\alpha)=\alpha$ (Lemma 3.2) and the limit profile

$$F(\alpha)=\frac{a_\alpha^2}{4}. \tag{7}$$

Throughout, C and c denote positive constants whose dependence is indicated, and K denotes a compact subset of $(0,\infty)$.

2.2. Classical inputs.

We use the following results as published; they are not reproved here.

Levin–Lubinsky [5]. For $n\ge1$ let $I_n=(c_n,d_n)$ with $-\infty\le c_n<d_n\le\infty$, and suppose that $[-r^*,r^*]\subset I_n$ for some $r^*>1$ and all n. Definition 1.1 [5, pp. 6–7] says that $\{Q_n\}\in\mathcal{Q}$ if the weights e^{-2nQ_n} on I_n satisfy five conditions. (i) $Q_n(x)/\log(2+|x|)\to\infty$ at c_n+ and d_n-. (ii) Q_n' is strictly increasing and continuous in I_n. (iii) There are $\alpha\in(0,1)$ and $C>0$ with $|Q_n'(x)-Q_n'(y)|\le C|x-y|^\alpha$ for $n\ge1$ and $x,y\in[-r^*,r^*]$. (iv) There are $\alpha_1\in(1/2,1)$, $C_1>0$ and an open neighbourhood I_0 of 1 and -1 with $|Q_n'(x)-Q_n'(y)|\le C_1|x-y|^{\alpha_1}$ for $n\ge1$ and $x,y\in I_n\cap I_0$. (v) $[-1,1]$ is the support of the equilibrium distribution of Q_n. By Remark (b) after that definition [5, p. 6], condition (v) is equivalent to the Mhaskar–Rakhmanov–Saff equations

$$\frac{1}{\pi}\int_{-1}^{1}\frac{xQ_n'(x)}{\sqrt{1-x^2}}\,\mathrm{d}x=1,\qquad \frac{1}{\pi}\int_{-1}^{1}\frac{Q_n'(x)}{\sqrt{1-x^2}}\,\mathrm{d}x=0. \tag{8}$$

Let $p_{n,m}$ be the orthonormal polynomial of degree m for e^{-2nQ_n} on I_n, with leading coefficient $\gamma_{n,m}$, and write the recurrence as in [5, p. 124],

$$xp_{n,m}(x)=A_{n,m}p_{n,m+1}(x)+B_{n,m}p_{n,m}(x)+A_{n,m-1}p_{n,m-1}(x),\qquad A_{n,m}=\frac{\gamma_{n,m}}{\gamma_{n,m+1}}, \tag{9}$$

the last identity being stated in the proof of Theorem 13.4 [5, p. 138]. (The recurrence printed in Theorem 2.3(c) [5, p. 12] contains a misprint, $A_{n,m}p_{n,m-1}$ for $A_{n,m}p_{n,m+1}$; we follow Chapter 13.) Theorem 13.4 [5, p. 124] states: if $\{Q_n\}\in\mathcal{Q}$ and $\varepsilon\in(0,1/3)$, then for some $\tau_1>0$, uniformly for $m=m(n)$ with $|m-n|\le n^{1/3-\varepsilon}$,

$$A_{n,m}=\frac{1}{2}+O\big(n^{-\tau_1}\big),\qquad B_{n,m}=O\big(n^{-\tau_1}\big). \tag{10}$$

Floater [3]. For a B-spline B of degree $d\ge2$ with simple knots, Lemma 2 and inequality (11) of [3], together with the remark following (11), give $BB''-\frac{d-1}{d}(B')^2<0$ on every interior knot interval, while

on the two end intervals B is a constant multiple of a dth power of the distance to the end knot. Since $BB'' - (B')^2 = \{BB'' - \frac{d-1}{d}(B')^2\} - \frac{1}{d}(B')^2$, this yields $(\log B)'' < 0$ on every open knot interval of the support. For w_M the degree is $d = M - 1$ and the knots are $-M/2, -M/2+1, \dots, M/2$.

3. The limit profile.

Lemma 3.1 *The function Λ' is odd and is an increasing homeomorphism of $\mathbb{R}$ onto $(-1/2, 1/2)$. For $s > 0$,*

$$0 < \Lambda''(s) \le \frac{1}{4}, \qquad 0 < \Lambda'(s) \le \frac{s}{4}, \qquad \frac{1}{2} - \frac{1}{s} < \Lambda'(s) \le \frac{1}{2} - \frac{1}{s+2}. \tag{11}$$

Consequently I' is odd, continuous and strictly increasing on $(-1/2, 1/2)$, and for $0 \le y \le y' < 1/2$,

$$I'(y) \ge 0, \qquad I'(y') - I'(y) \ge 4(y' - y), \qquad I'(y) \ge \frac{1}{1/2 - y} - 2. \tag{12}$$

Proof Oddness of (4) follows from $1/(e^{-s} - 1) = -1 - 1/(e^s - 1)$. With $u = s/2 > 0$, (4) gives $\Lambda'(s) = (u \cosh u - \sinh u)/(2u \sinh u)$ and $\Lambda''(s) = (\sinh^2 u - u^2)/(4u^2 \sinh^2 u)$. The functions $u \cosh u - \sinh u$ and $u^2 \sinh u - u \cosh u + \sinh u$ vanish at 0 and have derivatives $u \sinh u > 0$ and $u \sinh u + u^2 \cosh u \ge 0$, so

$$\sinh u < u \cosh u, \qquad u \cosh u - \sinh u \le u^2 \sinh u, \qquad u > 0. \tag{13}$$

The first inequality gives $\Lambda'(s) > 0$ and, squared with $\cosh^2 u = 1 + \sinh^2 u$, $\sinh^2 u - u^2 \le u^2 \sinh^2 u$, i.e. $\Lambda''(s) \le 1/4$; the second gives $\Lambda'(s) \le u/2 = s/4$; and $\sinh u > u$ gives $\Lambda''(s) > 0$. The bounds $e^s > 1 + s$ and $e^s \ge 1 + s + s^2/2$ give the two outer inequalities of (11). By the mean value theorem and oddness, Λ' is strictly increasing and $\Lambda'(\sigma') - \Lambda'(\sigma) \le (\sigma' - \sigma)/4$ for $0 \le \sigma \le \sigma'$; the bound $|\Lambda'(s)| \le |s|/4$ gives continuity at 0, and $\frac{1}{2} - \frac{1}{s} < \Lambda'(s) < \frac{1}{2}$ with the intermediate value theorem gives surjectivity onto $(-1/2, 1/2)$. The inverse I' of a strictly increasing continuous bijection is continuous and strictly increasing. For $\sigma = I'(y) \le \sigma' = I'(y')$ we have $y' - y = \Lambda'(\sigma') - \Lambda'(\sigma) \le (\sigma' - \sigma)/4$, and $y = \Lambda'(\sigma) \le \frac{1}{2} - \frac{1}{\sigma+2}$ gives the last bound in (12). □

Lemma 3.2 *The function H is continuous and strictly increasing on $[0, 1/2)$, $H(0) = 0$, and $H(a) \to \infty$ as $a \uparrow 1/2$. Hence a_α and $F(\alpha)$ are well defined for every $\alpha > 0$, $\alpha \mapsto a_\alpha$ is continuous and strictly increasing, and for $0 \le a \le a' \le b < 1/2$ and $0 < \mathrm{lo} \le a$,*

$$H(a') - H(a) \ge \mathrm{lo}\,(a' - a). \tag{14}$$

Proof By Lemma 3.1 the function $G(y) = yI'(y)$ is continuous on $[0, b]$, vanishes at 0 and is strictly increasing, and $G(y') - G(y) \ge 4y(y' - y)$ for $0 \le y \le y' \le b$. Continuity and strict monotonicity of $H = \mathcal{H}_G$ follow, the latter from the strict inequality at $\theta = \pi/2$. Integrating $G(a' \sin\theta) - G(a \sin\theta) \ge 4\,\mathrm{lo}\,(a' - a) \sin^2\theta$ with $\int_0^{\pi/2} \sin^2\theta \,\mathrm{d}\theta = \pi/4$ gives (14). For the divergence let $0 < \delta \le 1/8$ and $a = \frac{1}{2} - \frac{\delta^2}{4}$. For $\pi/2 - \delta \le \theta \le \pi/2$ one has $\sin\theta = \cos(\pi/2 - \theta) \ge 1 - \delta^2/2$, hence $y = a \sin\theta \ge 1/8$ and $\frac{1}{2} - y \le \delta^2/2$, and (12) gives

$G(y) \geq \frac{y}{1/2-y} - 2y \geq \frac{1}{4\delta^2} - 1$. Since $G \geq 0$,

$$\pi H\Big(\frac{1}{2} - \frac{\delta^2}{4}\Big) \geq \delta\Big(\frac{1}{4\delta^2} - 1\Big) = \frac{1}{4\delta} - \delta, \tag{15}$$

which tends to ∞ as $\delta \downarrow 0$; monotonicity completes the proof. □

4. A saddlepoint estimate with two derivatives.

Lemma 4.1 *For every $b \in (0, 1/2)$ there are M_0 and C such that for $M \geq M_0$*

$$\Big\| \log g_M + MI - \frac{1}{2}\log\frac{M}{2\pi v} - \frac{q}{M} \Big\|_{C^2([-b,b])} \leq \frac{C}{M^{3/2}}, \qquad q = \frac{\kappa_4}{8v^2} - \frac{5\kappa_3^2}{24v^3}. \tag{16}$$

In particular $\|\varphi_M' - I'\|_{C^1([-b,b])} \leq C/M$ and $\sup_{M\geq M_0}\sup_{|x|\leq b}|\varphi_M''(x)| < \infty$.

Proof *Representation.* For real s the probability density $e^{sy}w_M(y)A(s)^{-M}$ has characteristic function $\{A(s+iu)/A(s)\}^M$. Since $|2\sinh\{(s+iu)/2\}| \leq 2\cosh(s/2)$ and $|s+iu| \geq |u|$,

$$|A(s+iu)| \leq \frac{2\cosh(s/2)}{|u|}, \tag{17}$$

so for $M \geq 2$ the characteristic function is integrable and Fourier inversion holds pointwise. Taking $y = Mx$, $s = s(x)$, and using $A(s)^M e^{-sMx} = e^{M\{\Lambda(s)-sx\}} = e^{-MI(x)}$, we obtain for $|x| < 1/2$

$$g_M(x) = \frac{M}{2\pi}e^{-MI(x)}J_M(x), \qquad J_M(x) = \int_{\mathbb{R}} G(x,u)^M\,\mathrm{d}u, \qquad G(x,u) = e^{-iux}\frac{A(s(x)+iu)}{A(s(x))}. \tag{18}$$

Bounds on G. The map s is real-analytic, since $\Lambda'' > 0$, and $S = s([-b,b])$ is compact. As a characteristic function of a law with a density, $|G(x,u)| < 1$ for $u \neq 0$; by continuity, compactness and (17), for every $\varepsilon > 0$ there is $\rho_\varepsilon < 1$ with $|G| \leq \rho_\varepsilon$ for $|x| \leq b$ and $|u| \geq \varepsilon$, and $|G| \leq C/|u|$ for $|u| \geq 1$. Differentiating in x,

$$\partial_x G = -iuG + s'(x)\,e^{-iux}\,\frac{A'(s+iu)A(s) - A(s+iu)A'(s)}{A(s)^2}. \tag{19}$$

Because $A'(z) = \cosh(z/2)/z - 2\sinh(z/2)/z^2$ satisfies $|A'(s+iu)| \leq C/|u|$ for $|u| \geq 1$, the second term of (19) is $O(1/|u|)$ and $|uG| \leq C$, so $|\partial_x G| \leq C$. In $\partial_x^2 G$ the only term not bounded by a constant is $-iu\,\partial_x G = -u^2 G + O(1)$, so $|\partial_x^2 G| \leq C(1+|u|)$.

Outer region. For $j \leq 2$, $\partial_x^j(G^M)$ is a combination of $MG^{M-1}\partial_x^j G$ and $M(M-1)G^{M-2}(\partial_x G)^2$. Hence $|\partial_x^j(G^M)| \leq CM^2(1+|u|)|G|^{M-2}$, and for $|u| \geq \varepsilon$ and $M \geq 7$, $|G|^{M-2} \leq \rho_\varepsilon^{M-6}|G|^4$ with $(1+|u|)|G|^4$ integrable. This majorant, uniform in $|x| \leq b$, justifies differentiating J_M twice under the integral sign, and

$$\sup_{|x|\le b}\Big|\partial_x^j \int_{|u|\ge\varepsilon} G(x,u)^M\,\mathrm{d}u\Big| \le CM^2\rho_\varepsilon^{M-6}, \qquad j=0,1,2. \tag{20}$$

Inner region. Since $A>0$ on $\mathbb{R}$, we may choose $\varepsilon\in(0,1]$ so that $\Re A(\sigma+iu)>0$ for σ in a neighbourhood of S and $|u|\le\varepsilon$. There $G^M=e^{M\Psi}$ with the principal logarithm, $\Psi(x,u)=\Lambda(s(x)+iu)-\Lambda(s(x))-iux$, and Ψ is real-analytic in x and analytic in u. Taylor's formula, with Cauchy's estimates for the x-derivatives of the remainder, gives

$$\Psi=-\frac{v}{2}u^2-\frac{i\kappa_3}{6}u^3+\frac{\kappa_4}{24}u^4+\Psi_5, \qquad |\partial_x^j\Psi_5|\le C|u|^5, \qquad j\le 2,\ |u|\le\varepsilon. \tag{21}$$

Shrinking ε we also have $\Re\Psi\le -cu^2$. Moreover, $\partial_x\Psi=s'\{\Lambda'(s+iu)-\Lambda'(s)\}-iu$ has u-linear coefficient $i(s'v-1)$, and $\partial_x^2\Psi$ has u-linear coefficient $i(s''v+s'^2\kappa_3)$. Differentiating $\Lambda'(s(x))=x$ once and twice shows that both coefficients vanish, so

$$s'v=1, \qquad s''v+s'^2\kappa_3=0, \qquad |\partial_x\Psi|+|\partial_x^2\Psi|\le Cu^2, \qquad |u|\le\varepsilon. \tag{22}$$

Put $u=z/\sqrt{M}$ and $Z=M^{1/12}$. On $Z<|z|\le\varepsilon\sqrt{M}$, (22) gives $|\partial_x^j e^{M\Psi}|\le C(1+z^4)e^{-cz^2}$, whose integral is $O(e^{-cM^{1/6}/2})$. On $|z|\le Z$ write $M\Psi=-vz^2/2+w$, so that

$$w=-\frac{i\kappa_3 z^3}{6\sqrt{M}}+\frac{\kappa_4 z^4}{24M}+M\Psi_5\Big(x,\frac{z}{\sqrt{M}}\Big), \qquad |\partial_x^j w|\le\frac{C(1+|z|)^5}{\sqrt{M}}\le CM^{-1/12}, \qquad j\le 2. \tag{23}$$

For $E=e^w-1-w-w^2/2$ one has $\partial_x E=(e^w-1-w)\partial_x w$ and $\partial_x^2E=(e^w-1-w)\partial_x^2w+(e^w-1)(\partial_x w)^2$, so $|\partial_x^jE|\le C(1+|z|)^{15}M^{-3/2}$ for $j\le 2$. Collecting every term of order $M^{-3/2}$ into $\mathcal{E}_M$,

$$e^{M\Psi}=e^{-vz^2/2}\Big[1-\frac{i\kappa_3 z^3}{6\sqrt{M}}+\frac{1}{M}\Big(\frac{\kappa_4 z^4}{24}-\frac{\kappa_3^2 z^6}{72}\Big)+\mathcal{E}_M\Big], \qquad |\partial_x^j\mathcal{E}_M|\le\frac{C(1+|z|)^{18}}{M^{3/2}}. \tag{24}$$

The x-derivatives that fall on $e^{-vz^2/2}$ produce polynomial factors in z, so the remainder contributes $O(M^{-3/2})$ in C^2 after integration over $|z|\le Z$. The term in z^3 and its x-derivatives are odd in z and integrate to zero over the symmetric range, and completing the Gaussian moments to $\mathbb{R}$ costs $O(e^{-cZ^2})$. The moments $\int z^4e^{-vz^2/2}\,\mathrm{d}z=3\sqrt{2\pi}\,v^{-5/2}$ and $\int z^6e^{-vz^2/2}\,\mathrm{d}z=15\sqrt{2\pi}\,v^{-7/2}$, together with (20), give

$$J_M=\sqrt{\frac{2\pi}{Mv}}\Big(1+\frac{q}{M}+R_M\Big), \qquad \|R_M\|_{C^2([-b,b])}\le\frac{C}{M^{3/2}}. \tag{25}$$

By (18), $\log g_M+MI-\frac12\log\{M/(2\pi v)\}=\log(1+q/M+R_M)$, and since q is smooth on $[-b,b]$ this equals $q/M+O(M^{-3/2})$ in $C^2([-b,b])$. Dividing by $-M$ and differentiating gives the last assertions, because $\varphi_M'-I'=-M^{-1}\{\frac12\log(1/v)\}'+O(M^{-2})$ in C^1. □

5. Equilibrium endpoints.

Lemma 5.1 *For $M \geq 3$ the function φ'_M is odd, continuous and strictly increasing on $(-1/2, 1/2)$, with $\varphi'_M(0) = 0$; for $M = 2$, $x\varphi'_2(x) = |x|/(1-2|x|)$; for $M = 1$, $\varphi'_1 \equiv 0$. For $M \geq 2$ and $x \uparrow 1/2$, $x\varphi'_M(x) \geq c_M/(1/2 - x)$.*

Proof The density w_M is even and positive on $(-M/2, M/2)$, belongs to C^{M-2}, and on each knot interval is a polynomial of degree $M-1$. For $M \geq 3$ it is C^1, and by Floater's result recalled in Section 2.2, $(\log w_M)'' < 0$ on every open knot interval; hence $(\log w_M)'$ is strictly decreasing on $(-M/2, M/2)$, and the same holds for $\log g_M$, so φ'_M is strictly increasing. For $M = 2$, $g_2(x) = 2(1-2|x|)$. On the last knot interval $g_M(x)$ is a positive multiple of $(1/2 - x)^{M-1}$, so $\varphi'_M(x) = (M-1)/\{M(1/2-x)\}$ there. □

Lemma 5.2 (a) *For every $M \geq 2$ and $t > 0$ the equation $H_M(a) = t$ has a unique root $a_{t,M} \in (0, 1/2)$; for $M = 1$ it has no root.* (b) *For every compact $K \subset (0, \infty)$ there are $0 < \mathrm{lo} < b < 1/2$, C_K and M_K such that for $M \geq M_K$ and $t \in K$ both roots a_t and $a_{t,M}$ lie in (lo, b) and*

$$|a_{t,M} - a_t| \leq \frac{C_K}{M}. \tag{26}$$

Proof (a) By Lemma 5.1, $x\varphi'_M(x)$ is continuous on $[0, 1/2)$, vanishes at 0, is strictly increasing and is bounded below by $c_M/(1/2-x)$ near $1/2$; as in Lemma 3.2, H_M increases continuously from 0 to ∞. For $M = 1$, $H_1 \equiv 0$. (b) Let $K \subset [t_1, t_2]$. By Lemma 3.2 choose $b < 1/2$ with $H(b) \geq t_2 + 1$, and $\mathrm{lo} \in (0, b)$ with $H(\mathrm{lo}) < t_1/2$. By Lemma 4.1, $|\varphi'_M - I'| \leq C/M$ on $[0, b]$, hence $|x\varphi'_M(x) - xI'(x)| \leq bC/M$ and

$$|H_M(a) - H(a)| \leq \frac{bC}{2M}, \qquad 0 \leq a \leq b. \tag{27}$$

For $M > bC(1/t_1 + 1)$ this gives $H_M(\mathrm{lo}) < t_1 \leq t \leq t_2 < H_M(b)$, so $a_{t,M} \in (\mathrm{lo}, b)$, and likewise $a_t \in (\mathrm{lo}, b)$. Then $|H(a_{t,M}) - H(a_t)| = |H(a_{t,M}) - H_M(a_{t,M})| \leq bC/(2M)$, and (14) gives (26) with $C_K = bC/(2\,\mathrm{lo})$. □

6. The Levin–Lubinsky class.

For $N, M \in \mathbb{N}$ with $M \geq 2$ put $t = N/M$, $a = a_{N,M} = a_{t,M}$, and define $Q_{N,M}$ on $I_{N,M} = (-1/(2a), 1/(2a))$ by

$$e^{-2NQ_{N,M}(y)} = a\, g_M(ay), \qquad Q_{N,M}(y) = \frac{M}{2N}\varphi_M(ay) - \frac{\log a}{2N}, \qquad Q'_{N,M}(y) = \frac{aM}{2N}\varphi'_M(ay). \tag{28}$$

Lemma 6.1 *Let $K \subset (0, \infty)$ be compact. There are N_0, $r^* > 1$, $C > 0$, $C_1 > 0$ and an open neighbourhood I_0 of ± 1 such that, with $\alpha = 1/4$ and $\alpha_1 = 3/4$, the weights (28) with $N \geq N_0$ and $N/M \in K$ satisfy conditions* (i)–(v) *of Definition* 1.1 *of* [5] *with these constants.*

Proof Let b, lo be as in Lemma 5.2(b) and choose $r^* > 1$ with $r^* b < 1/2$, which is possible because $b < 1/2$. For $N \geq N_0$ we have $M \geq M_K$, $a \in (\mathrm{lo}, b)$ and so $[-r^*, r^*] \subset I_{N,M}$. Condition (i) holds because g_M vanishes at $\pm 1/2$, so $Q_{N,M} \to \infty$ at the finite endpoints of $I_{N,M}$. Condition (ii) is Lemma 5.1, since $M \geq 3$. By (28)

and Lemma 4.1 applied on $[-r^*b, r^*b]$, for $x, y \in [-r^*, r^*]$,

$$|Q'_{N,M}(x) - Q'_{N,M}(y)| \le \frac{a^2 M}{2N}\Big(\sup_{|u|\le r^* b} |\varphi''_M(u)|\Big)|x-y| \le L|x-y|, \tag{29}$$

with L independent of (N, M), because $M/N \le 1/\min K$. A Lipschitz bound on $[-r^*, r^*]$ implies $|Q'(x) - Q'(y)| \le L(2r^*)^{1-\gamma}|x-y|^\gamma$ for every $\gamma \in (0,1]$, which gives (iii) with $\alpha = 1/4$ and (iv) with $\alpha_1 = 3/4$ and $I_0 = \{x : \big||x|-1\big| < \eta\}$, $1+\eta < r^*$; note $\alpha \le \alpha_1 - 1/2$, as in Remark (e) after Definition 1.1. For (v) we use (8). The second equation holds because $Q'_{N,M}$ is odd. In the first, the substitution $x = \cos\theta$ and then $x = a\sin\theta$ in (28), with the oddness of φ'_M, give

$$\frac{1}{\pi}\int_{-1}^{1} \frac{xQ'_{N,M}(x)}{\sqrt{1-x^2}}\,\mathrm{d}x = \frac{1}{\pi}\int_0^\pi \cos\theta\, Q'_{N,M}(\cos\theta)\,\mathrm{d}\theta = \frac{M}{N}\,H_M(a), \tag{30}$$

which equals 1 exactly when $H_M(a) = N/M$, the defining equation of $a = a_{N,M}$. □

7. Proof of the theorem.

Theorem 7.1 *For every compact $K \subset (0,\infty)$,*

$$\lim_{L\to\infty} \sup_{\substack{N,M\in\mathbb{N},\ N\ge L\\ N/M\in K}} \Big|\frac{\beta_N^{[M]}}{M^2} - F\Big(\frac{N}{M}\Big)\Big| = 0. \tag{31}$$

Proof *Scaling.* Let $y = Y_M/a$ with $a = a_{N,M}$, so that $S_M = May$ and y has the weight $e^{-2NQ_{N,M}}$ of (28). If h_n^y and h_n^S are the squared norms of the monic orthogonal polynomials of y and of S_M, then $P_n^{[M]}(X) = (Ma)^n P_n^y\{X/(Ma)\}$ and $h_n^S = (Ma)^{2n}h_n^y$. Since $\gamma_{N,n} = (h_n^y)^{-1/2}$, (9) gives

$$\beta_N^{[M]} = \frac{h_N^S}{h_{N-1}^S} = (Ma)^2\frac{h_N^y}{h_{N-1}^y} = (Ma)^2 A_{N,N-1}^2. \tag{32}$$

Sequences. Let $K \subset [t_1, t_2]$ with $0 < t_1 < t_2$ and $E(N,M) = \beta_N^{[M]}/M^2 - F(N/M)$. Let $m : \mathbb{N} \to \mathbb{N}$ satisfy $n/m(n) \in [t_1, t_2]$ for all large n. Put $Q_n = Q_{n,m(n)}$ for $n \ge N_0$, with N_0 from Lemma 6.1 for $[t_1, t_2]$, and $Q_n(x) = x^2$ on $I_n = \mathbb{R}$ for $n < N_0$; the latter satisfies (i)–(v), since $\frac{1}{\pi}\int_{-1}^{1} 2x^2(1-x^2)^{-1/2}\,\mathrm{d}x = 1$. Then $\{Q_n\} \in \mathcal{Q}$, and (10) with $m = n-1$, which lies in the window $|m-n| \le n^{1/3-\varepsilon}$, gives $A_{n,n-1} \to 1/2$. By (32), $E(n, m(n)) = a_{n,m(n)}^2 A_{n,n-1}^2 - a_t^2/4$ with $t = n/m(n)$, and Lemma 5.2(b) gives $a_{n,m(n)} - a_t \to 0$. Hence $E(n, m(n)) \to 0$.

Uniformity. Suppose (31) fails. Then there are $\varepsilon > 0$ and pairs (N_j, M_j) with N_j strictly increasing, $N_j/M_j \in K$ and $|E(N_j, M_j)| \ge \varepsilon$. For every large n there is M with $n/M \in [t_1, t_2]$, for instance $M = \lceil n/t_2 \rceil$.

Define $m(N_j) = M_j$ and fill the other large n in this way. By the previous step $E(n, m(n)) \to 0$, which contradicts $|E(N_j, M_j)| \geq \varepsilon$. □

8. The profile near 0 and ∞.

Corollary 8.1 *The profile F is continuous and strictly increasing on $(0, \infty)$, $0 < F < 1/16$, $F(\alpha) \to 1/16$ as $\alpha \to \infty$, and as $\alpha \downarrow 0$*

$$F(\alpha) = \frac{\alpha}{12} - \frac{\alpha^2}{20} + \frac{4\alpha^3}{525} + \frac{2\alpha^4}{525} + \frac{368\alpha^5}{336875} - \frac{544\alpha^6}{1340625} + O(\alpha^7). \tag{33}$$

Proof The first assertions follow from Lemma 3.2, since $a_\alpha \uparrow 1/2$. The function Λ' is analytic at 0 with $\Lambda'(s) = \sum_{k \geq 1} B_{2k} s^{2k-1}/(2k)! = s/12 - s^3/720 + \cdots$, where B_{2k} are the Bernoulli numbers, and $\Lambda''(0) = 1/12 \neq 0$. Hence I' is analytic and odd at 0, and inversion of the series gives

$$I'(x) = 12x + \frac{144}{5}x^3 + \frac{19008}{175}x^5 + \frac{393984}{875}x^7 + O(x^9). \tag{34}$$

Inserting this in (6) and using $\int_0^{\pi/2} \sin^{2k}\theta \, \mathrm{d}\theta = \frac{\pi}{2}\binom{2k}{k}4^{-k}$ gives

$$H(a) = 3a^2 + \frac{27}{5}a^4 + \frac{594}{35}a^6 + \frac{1539}{25}a^8 + O(a^{10}), \tag{35}$$

and inverting $\alpha = H(a)$ for a^2 to sixth order gives (33); the coefficients were obtained by exact rational arithmetic [1]. □

Milovanović's Conjecture 2 [6, p. 347] asserts that for fixed $k \geq 1$ and $m \to \infty$

$$\beta_k^{[m]} \asymp \frac{k}{12}m - \frac{k(k-1)}{20} + \frac{4k(k-1)(k-2)}{525m} + \frac{2k(k-1)(k-2)(5k-9)}{2625m^2} + \frac{2k(k-1)(k-2)(552k^2 - 1279k + 255)}{1010625m^3} + \cdots. \tag{36}$$

The coefficients of the leading-degree monomials $k^j m^{2-j}$ of the five terms of (36) are $\frac{1}{12}$, $-\frac{1}{20}$, $\frac{4}{525}$, $\frac{10}{2625} = \frac{2}{525}$ and $\frac{1104}{1010625} = \frac{368}{336875}$, which are exactly the first five coefficients of (33). This is a consistency between two different regimes, not a proof of the conjecture or a consequence of Theorem 7.1, because no uniformity connecting fixed k with $k \asymp m$ is available. The fixed-degree first two terms are the content of Theorem 5.1 of the manuscript V035, which Theorem 7.1 neither implies nor uses; there the subleading term $+k/20$ of $-k(k-1)/20$ is visible, while in the ratio regime it is of lower order.

9. Certification and numerical evidence.

Lean 4. The library `UniformLanczos` [1], compiled with the Lean 4 toolchain `v4.35.0-rc3` and Mathlib commit `0272bbe1c093`, contains no `sorry`, and every main declaration depends only on the axioms `propext`,

`Classical.choice` and `Quot.sound`. It certifies the following statements. Lemma 3.1 in full, with Λ' in the closed form (4), including the inverse I' and (12) (module `Lambda`). Lemma 3.2 in full, with H in the θ-form of (6) (`Endpoint`, `EndpointConcrete`). Lemma 5.2(b) relative to two hypotheses on the finite-M functions, namely the bound $|\varphi'_M - I'| \le C/M$ of Lemma 4.1 and the regularity of Lemma 5.1 (`endpoint_concrete`). The cancellations (22) and the identities (5) (`Cancellation`). In Lemma 6.1, the formula (28) for Q', the Lipschitz transfer (29), Lipschitz $\Rightarrow$ Hölder, $[-r^*, r^*] \subset I_{N,M}$, the arcsine substitution in (30) for Lebesgue integrals without any integrability hypothesis, and the equivalence of (8) with the endpoint equation (`ClassQ`, `MRS`, `Arcsine`). The scaling (32) and the uniformity argument with the filling step (`Diagonal`). The first two terms of (33), unconditionally: the bound $|I'(x) - 12x - \frac{144}{5}x^3| \le 520x^5$ for $0 \le x \le \Lambda'(1)$, obtained from $|\Lambda'(s) - \frac{s}{12} + \frac{s^3}{720}| \le s^5/25600$ for $0 < s \le 2$, and the resulting $|F(\alpha) - \frac{\alpha}{12} + \frac{\alpha^2}{20}| \le K\alpha^3$ (`Corollary`, `Expansion`). Not formalised are the Fourier–Laplace analysis of Lemma 4.1, Lemma 5.1 (Floater's inequality), the coefficients of (33) beyond the second, which rest on exact rational series arithmetic, and the potential theory of [5] used in (8) and (10); in the formal statements these enter as explicit hypotheses.

Numerical checks. Table 1 compares $\beta_N^{[M]}/M^2$ with $F(N/M)$. The coefficients were computed by Gauss–Legendre discretisation on each knot interval, which is exact for the first N coefficients, and the Stieltjes procedure in 64-bit-mantissa arithmetic; they agree to about 17 digits with exact rational values for small (N, M) [1]. The quantity ME is essentially independent of M, which suggests $E = c(t)/M + O(M^{-2})$; this rate is not claimed. The same programs confirm (16): at $x = 0.25$, M times the left side of (16) without the term q/M tends to $q(0.25) \approx -0.184391$.

Table 1. The error $E = \beta_N^{[M]}/M^2 - F(N/M)$ for $N = tM$; $F(t)$ to seven digits.

t	$F(t)$	ME, $M = 100$	ME, $M = 200$	ME, $M = 400$	E, $M = 400$
1/4	0.0178432	0.010828	0.010800	0.010786	2.70×10^{-5}
1/2	0.0303768	0.017171	0.017111	0.017081	4.27×10^{-5}
1	0.0445361	0.017648	0.017602	0.017579	4.39×10^{-5}
2	0.0548091	0.010479	0.010471	0.010466	2.62×10^{-5}
4	0.0598372	0.004340	0.004341	0.004342	1.09×10^{-5}

Status. With the published inputs of Section 2.2, the proof above is complete. This revision was prepared with the assistance of Claude (Anthropic) as a technical tool, for checking the proofs against the cited sources, completing the missing estimates, writing the Lean 4 certificates and running the computations; the decision on incorporation into the manuscript rests with the author.

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